

Response to request for information (RFI) on the Development of an Artificial Intelligence Action Plan; Submitted via email to [REDACTED]**March 15, 2025**

We appreciate the opportunity to offer comments as the Trump Administration develops an Artificial Intelligence (AI) Action Plan to ensure the United States remains at the forefront of AI innovation. As an applied safety science leader for over 130 years, we understand the role safety plays in supporting innovation. Through testing, inspection, certification, and advisory services, UL Solutions helps verify products meet safety, security, and sustainability standards. As AI is becoming an increasingly integral part of the software ecosystem and being relied upon by products with safety critical functions, we have helped examine the safety, security, and sustainability impacts of AI in autonomous systems, healthcare applications and other technologies. This work informs our recommendation below on the need for development of a comprehensive AI standard that fosters innovation and advancement and at the same time promotes safe AI, supports responsible AI development, and aligns with existing international standards from ISO/IEC.

Comprehensive AI Standard for Safe & Responsible AI Development

The rapid advancement and integration of AI into critical sectors such as healthcare, transportation, manufacturing, and defense necessitate a structured approach to ensure AI technologies remain not only beneficial but also safe and trustworthy. A comprehensive AI standard can foster innovation and advancement, while promoting safety and supporting responsible development. Without a strong foundational standard, AI systems may pose risks such as bias, lack of transparency, security vulnerabilities, and system failures, which could have severe consequences for public safety, economic stability, and trust in AI technologies.

By integrating risk-based safety principles, the proposed AI standard can help meet the objectives of Executive Order 14179 (Removing Barriers to American Leadership in Artificial Intelligence) while ensuring that AI is safe for users, ethically aligned, and resilient against misuse. Key areas that should be included in the standard, are:

- **Quality Management:** Ensuring structured development and testing processes.
- **Robustness & Reliability:** Addressing AI system performance across different conditions.
- **Transparency & Explainability:** Enabling users and regulators to understand AI decisions.

- Control & Oversight: Providing mechanisms for human intervention.
- Fairness & Bias: Ensuring equitable outcomes and reducing discrimination.
- Data Privacy & Security: Implementing safeguards to protect sensitive data.
- AI-Based Functional Safety: Ensuring AI components in safety-critical systems function predictably.
- Lifecycle Management: Establishing guidelines for AI system updates, maintenance, and decommissioning.

A comprehensive AI standard should incorporate key safety principles and guidelines and provide a framework for assessing AI safety through technical, ethical, and governance-related principles. These are critical for ensuring AI systems are reliable, fair, and well-documented.

Executive Order 14179 emphasizes AI leadership by ensuring that innovation is not stifled by unnecessary regulations. A well-structured AI standard can balance innovation with safety by providing clear guidelines that enable businesses to develop and deploy AI responsibly while maintaining global competitiveness. A standardized safety framework provides the predictability, trust, and regulatory clarity that businesses need to accelerate AI adoption without facing inconsistent or burdensome regulatory requirements. A structured approach can streamline compliance and reduce uncertainty, ultimately enhancing US leadership in AI innovation.

A comprehensive AI standard should align with international best practices and existing AI safety frameworks such as those developed by UL Solutions, ISO/IEC JTC 1/SC 42, IEEE, and NIST. Many of these organizations have already established AI governance, risk management, and safety guidelines, including:

- UL: Outline of Investigation for Safety of AI-Based Products
- ISO/IEC 42001 (AI Management System)
- ISO/IEC 23894 (AI Risk Management)
- ISO/IEC TR 5469 (AI Functional Safety)
- NIST AI Risk Management Framework (AI RMF 1.0)

By adopting and integrating these internationally recognized best practices, the U.S. can ensure interoperability, facilitate global AI trade, and reinforce American leadership in AI policy and governance.

Applicable Standards in Healthcare

Given the critical need for safety and security in healthcare, we also suggest that the following standards should be a part of US medical and healthcare AI policy making:

- AAMI TIR 34971:2023; Application of ISO 14971 to machine learning in artificial intelligence—Guide
- UL 2900-2-1 Ed. 1-2017 Software Cybersecurity for Network-Connectable Products, Part 2-1: Particular Requirements for Network Connectable Components of Healthcare and Wellness Systems
- IEC 81001-5-1 Health Software and Health IT Systems Safety, Effectiveness and Security - Part 5-1: Security - Activities in The Product Life Cycle
- IEC 82304-1 Health Software - Part 1: General Requirements for Product Safety

Thank you again for the opportunity to provide comments as the Administration develops an AI Action Plan. Leveraging key safety principles and guidelines will enable trust in AI systems, drive the adoption of AI, and maintain US leadership in the global AI landscape. If you have further questions, please contact Derek Greenauer, Director, Public Affairs, Americas Region ([REDACTED])

Sincerely,

/s

Derek Greenauer
Director, Public Affairs, Americas Region
UL Solutions